

An-Najah National University
Master of AI
IoT Principles & Design
Course Project- Final Phase
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Deadline: 25/12/2025

Objective:

Following what you have developed in the first phase, the objective of this phase is involve data analytics for the data collected from the sensors in the developed IoT system. Furthermore, a proper Machine Learning (ML) model should be built and modified whenever needed. This ML model should guide the application control to make proper decisions (instead of having just rule-based decisions).

Requirements:

1. Modify the system you have implemented in phase#1 to perform the following:
 - a. Include additional types of sensors (in the attached document).
 - b. Designate one of the sensors to do initial filtration/processing before the data reaches the gateway. That is, this sensor represents a tier between the sensors and the gateway which, in turn, should send the data to the cloud.
 - c. Write a program to convert the data stored in the data warehouse into a form that is ready to be further analyzed with descriptive analysis (Spreadsheet, visualized form, etc,..)
 - d. Perform proper prescriptive analysis to the data based on the resultant descriptive analysis. Build a Machine Learning (ML) model for your system based on the data. Be careful that this model should be modified whenever needed.
 - e. Use the built ML model to help the control application making proper decisions so that proper commands are sent to the actuators (Prescriptive analysis).

Deliverables:

1. Write a paper that follows the IEEE double-column format that describes your design. This paper should contain the following:
 - a. Abstract and Introduction.
 - b. Literature review of the previous work.
 - c. Your proposed design (components, connections, and used protocols).
 - d. Simulation results that contain screenshots of the output.

e. Design evaluation: Your design should be evaluated in terms of correctness, availability, as well as the response time to process different events. Furthermore, the impact of using another ML model should be included in the evaluation to have final recommendations.

f. Conclusions and Future works.

2. Your source code. It should be clear and well-documented.

3. After the deadline, you have to do a presentation for your work. You have to present your design, simulation results, evaluation metrics, and recommendations. Furthermore, you will be requested to have a demo for your work.

4. Submit your work within the deadline stated above.